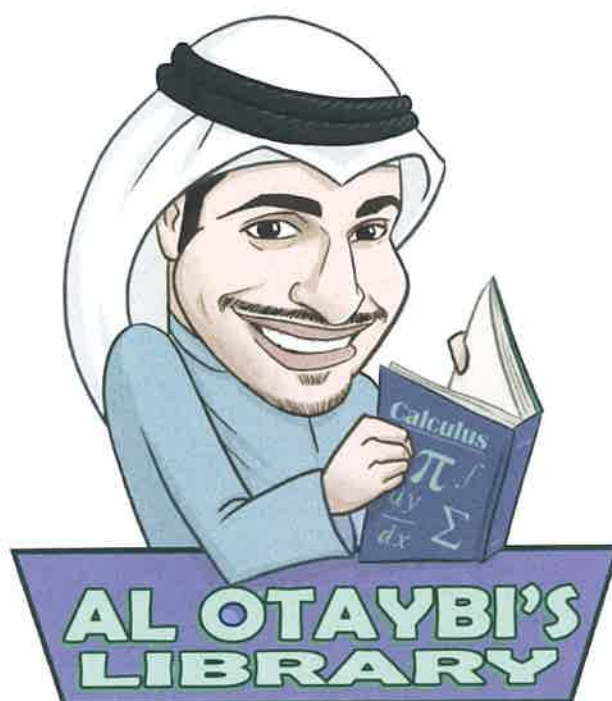


Kuwait University

2021/2022

Fall course



Calculus A

First

Sections : 2.4 – 2.5

2.4 The Precise Definition of a Limit

2.5 Continuity

(7)

2.4 The Precise Definition of a Limit

Use definition of limit to show that

$$\lim_{x \rightarrow 3} (4x - 5) = 7$$

Given any $\varepsilon > 0$ we want to find $\delta > 0$ such that:

$$0 < |x - 3| < \delta \iff |4x - 5 - 7| < \varepsilon$$

$$\iff |4x - 12| < \varepsilon$$

$$\iff 4|x - 3| < \varepsilon$$

$$\iff |x - 3| < \frac{\varepsilon}{4}$$

$$\therefore \text{choose } \delta = \frac{\varepsilon}{4}$$

Now

$$|x - 3| < \delta$$

$$|x - 3| < \frac{\varepsilon}{4}$$

$$4|x - 3| < \varepsilon$$

$$|4(x - 3)| < \varepsilon$$

$$|4x - 12| < \varepsilon$$

$$|4x - 5 - 7| < \varepsilon$$

Use definition of limit to show that

$$\lim_{x \rightarrow 1} (4 - 2x) = 2$$

Given any $\varepsilon > 0$ we want to find $\delta > 0$ such that

$$\begin{aligned} 0 < |x - 1| < \delta &\iff |4 - 2x - 2| < \varepsilon \\ &\iff |2 - 2x| < \varepsilon \\ &\iff 2|1 - x| < \varepsilon \\ &\iff |1 - x| < \frac{\varepsilon}{2} \\ &\iff |x - 1| < \frac{\varepsilon}{2} \end{aligned}$$

Now $\therefore$ choose $\delta = \frac{\varepsilon}{2}$

$$\begin{aligned} |x - 1| < \delta \\ |x - 1| < \frac{\varepsilon}{2} \\ |1 - x| < \frac{\varepsilon}{2} \\ 2|1 - x| < \varepsilon \\ |2 - 2x| < \varepsilon \\ |4 - 2x - 2| < \varepsilon \end{aligned}$$

Use ε and δ to prove that

$$\lim_{x \rightarrow a} (bx + c) = ba + c$$

Given any $\varepsilon > 0$ we want to find $\delta > 0$ such that

$$0 < |x - a| < \delta \iff |bx + c - ba - c| < \varepsilon$$

$$\iff |bx - ba| < \varepsilon$$

$$\iff b|x - a| < \varepsilon$$

$$\iff |x - a| < \frac{\varepsilon}{b}$$

$$\therefore \text{choose } \delta = \frac{\varepsilon}{b}$$

Now

$$|x - a| < \delta$$

$$|x - a| < \frac{\varepsilon}{b}$$

$$b|x - a| < \varepsilon$$

$$|bx - ba| < \varepsilon$$

$$|bx + c - ba - c| < \varepsilon$$



Text book question



Use definition of limit to show that

$$\lim_{x \rightarrow -2} \left(\frac{1}{2}x + 3 \right) = 2$$

Given any $\varepsilon > 0$ we want to find $\delta > 0$ such that

$$0 < |x + 2| < \delta \iff \left| \frac{1}{2}x + 3 - 2 \right| < \varepsilon$$

$$\iff \left| \frac{1}{2}x + 1 \right| < \varepsilon$$

$$\iff |x + 2| < 2\varepsilon$$

$$\therefore \text{choose } \delta = 2\varepsilon$$

Now

$$|x + 2| < \delta$$

$$|x + 2| < 2\varepsilon$$

$$\left| \frac{1}{2}x + 1 \right| < \varepsilon$$

$$\left| \frac{1}{2}x + 3 - 2 \right| < \varepsilon$$



Text book question



Prove the statement using the ε , δ definition of a limit

$$\lim_{x \rightarrow 10} \left(3 - \frac{4}{5}x \right) = -5$$

Given any $\varepsilon > 0$ we want to find $\delta > 0$ such that

$$0 < |x - 10| < \delta \iff \left| 3 - \frac{4}{5}x + 5 \right| < \varepsilon$$

$$\iff \left| 8 - \frac{4}{5}x \right| < \varepsilon$$

$$\iff \left| 40 - 4x \right| < 5\varepsilon$$

$$\iff 4 \left| 10 - x \right| < 5\varepsilon$$

$$\iff \left| 10 - x \right| < \frac{5}{4}\varepsilon$$

Now

$$\left| x - 10 \right| < \delta$$

$$\left| 10 - x \right| < \frac{5}{4}\varepsilon$$

$$4 \left| 10 - x \right| < 5\varepsilon$$

$$\left| 40 - 4x \right| < 5\varepsilon$$

$$\left| 8 - \frac{4}{5}x \right| < \varepsilon$$

$$\left| 3 - \frac{4}{5}x + 5 \right| < \varepsilon$$

$\therefore$ Choose $\delta = \frac{5}{4}\varepsilon$



Text book question



Prove the statement using the ε , δ definition of a limit

$$\lim_{x \rightarrow 1} \frac{2 + 4x}{3} = 2$$

Given any $\varepsilon > 0$ we want to find $\delta > 0$ such that

$$\begin{aligned} 0 < |x - 1| < \delta &\iff \left| \frac{2 + 4x}{3} - 2 \right| < \varepsilon \\ &\iff |2 + 4x - 6| < 3\varepsilon \\ &\iff |4x - 4| < 3\varepsilon \\ &\iff 4|x - 1| < 3\varepsilon \\ &\iff |x - 1| < \frac{3}{4}\varepsilon \end{aligned}$$

Now

$$\begin{aligned} |x - 1| &< \delta \\ |x - 1| &< \frac{3}{4}\varepsilon \\ 4|x - 1| &< 3\varepsilon \\ |4x - 4| &< 3\varepsilon \\ |2 + 4x - 6| &< 3\varepsilon \\ \left| \frac{2 + 4x}{3} - 2 \right| &< \varepsilon \end{aligned}$$

$\therefore$ Choose $\delta = \frac{3}{4}\varepsilon$

Use the definition of limit to show that

$$\lim_{x \rightarrow 3} (x - 1) = 2$$

Mar 2011

Given any $\varepsilon > 0$ we want to find $\delta > 0$ such that :

$$0 < |x - 3| < \delta \iff |x - 1 - 2| < \varepsilon$$
$$\iff |x - 3| < \varepsilon$$

$\therefore$ choose $\delta = \varepsilon$

Now

$$|x - 3| < \delta$$
$$|x - 3| < \varepsilon$$
$$|x - 1 - 2| < \varepsilon$$

Use the definition of limit to show that

$$\lim_{x \rightarrow 6} \left(9 - \frac{1}{6}x \right) = 8$$

Oct 2010

Given any $\varepsilon > 0$ we want to find $\delta > 0$ such that

$$0 < |x - 6| < \delta \iff \left| 9 - \frac{1}{6}x - 8 \right| < \varepsilon$$
$$\iff \left| 1 - \frac{1}{6}x \right| < \varepsilon$$
$$\iff |6 - x| < 6\varepsilon$$
$$\iff |x - 6| < 6\varepsilon$$

$\therefore$ choose $\delta = 6\varepsilon$

Now

$$|x - 6| < \delta$$
$$|x - 6| < 6\varepsilon$$
$$|6 - x| < 6\varepsilon$$
$$\left| 1 - \frac{1}{6}x \right| < \varepsilon$$
$$\left| 9 - \frac{1}{6}x - 8 \right| < \varepsilon$$

2.5 Continuity



Continuity at a number $x = a$



لاثبت أن الدالة متصلة عند رقم $x = a$

يجب تحقق الثلاث شروط التالية

1) $f(a)$ is exist

2) $\lim_{x \rightarrow a} f(x)$ is exist

3) $f(a) = \lim_{x \rightarrow a} f(x)$

Example

$$\text{Let } f(x) = \begin{cases} \frac{|x-3|}{x-3}, & x \neq 3 \\ 1, & x = 3 \end{cases}$$

Check for continuity at $x = 3$

1) $f(3) = 1$

2) $\lim_{x \rightarrow 3} \frac{|x-3|}{x-3}$

$$\begin{array}{l|l} \lim_{x \rightarrow 3^+} \frac{+(x-3)}{x-3} & \lim_{x \rightarrow 3^-} \frac{-(x-3)}{x-3} \\ \hline \lim_{x \rightarrow 3^+} 1 = 1 & \lim_{x \rightarrow 3^-} -1 = -1 \end{array}$$

$$\therefore \lim_{x \rightarrow 3^+} f(x) \neq \lim_{x \rightarrow 3^-} f(x) \longrightarrow \therefore \lim_{x \rightarrow 3} f(x) \text{ DNE}$$

$\Rightarrow \therefore f$ is Not continuous at $x=3$



Text book questions



Explain why the function is discontinuous at the given number a .

$$f(x) = \frac{1}{x+2}$$

$$a = -2$$

$\therefore f(-2)$ undefined

$\therefore$ at $x = -2$ $f(x)$ is not continuous

Explain why the function is discontinuous at the given number a .

$$f(x) = \begin{cases} \frac{x^2 - x}{x^2 - 1}, & \text{if } x \neq 1 \\ 1, & \text{if } x = 1 \end{cases} \quad a = 1$$

$$f(1) = 1$$

$$\lim_{x \rightarrow 1} \frac{x^2 - x}{x^2 - 1} = \lim_{x \rightarrow 1} \frac{x(x-1)}{(x-1)(x+1)} = \frac{1}{2}$$

$$\therefore f(1) \neq \lim_{x \rightarrow 1} f(x)$$

$\therefore$ at $x = 1$ $f(x)$ is not continuous

Explain why the function is discontinuous at the given number a .

$$f(x) = \begin{cases} \frac{1}{x+2}, & \text{if } x \neq -2 \\ 1, & \text{if } x = -2 \end{cases} \quad a = -2$$

$$f(-2) = 1$$

$$\lim_{x \rightarrow -2} \frac{1}{x+2}$$

$$\lim_{x \rightarrow -2^+} \frac{1}{x+2} = \frac{1}{0} = \infty$$

$$\lim_{x \rightarrow -2}$$

$$\lim_{x \rightarrow -2^-} \frac{1}{x+2} = \frac{1}{0^-} = -\infty$$

$$\therefore \lim_{x \rightarrow -2^+} f(x) \neq \lim_{x \rightarrow -2^-} f(x)$$

$$\therefore \lim_{x \rightarrow -2} f(x) \text{ DNE}$$

$\therefore$ at $x = -2$ is not continuous

Determine whether f is continuous at $x = 1$ where

$$f(x) = \begin{cases} 2^x, & \text{if } x \geq 1 \\ \frac{1-x^2}{|1-x|}, & \text{if } x < 1 \end{cases}$$

April 2017

1) $f(1) = 2^1 = 2$

2) $\lim_{x \rightarrow 1^+} 2^x = 2^1 = 2$

$$\lim_{x \rightarrow 1^-} \frac{1-x^2}{|1-x|}$$

$$\lim_{x \rightarrow 1^-} \frac{1-x^2}{+(1-x)}$$

$$\lim_{x \rightarrow 1^-} \frac{(1-x)(1+x)}{1-x} = 1+1 = 2$$

$$\therefore f(1) = \lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^-} f(x)$$

$\therefore$ at $x=1$ $f(x)$ is continuous.

Suppose that $f(x) = \begin{cases} \frac{\sqrt{6x-5} - \sqrt{3x-2}}{x^2 + 4x - 5} & , \text{ if } x > 1 \\ \frac{1}{4} & , \text{ if } x = 1 \\ \frac{|x-1|}{x^2 - 3x + 2} - \frac{3}{4} & , \text{ if } x < 1 \end{cases}$

Check for continuity at $x = 1$

past exam

1) $f(1) = \boxed{\frac{1}{4}}$

2) $\lim_{x \rightarrow 1^+} \frac{\sqrt{6x-5} - \sqrt{3x-2}}{x^2 + 4x - 5} \cdot \frac{\sqrt{6x-5} + \sqrt{3x-2}}{\sqrt{6x-5} + \sqrt{3x-2}}$

$= \lim_{x \rightarrow 1^+} \frac{6x-5 - (3x-2)}{(x^2 + 4x - 5)(\sqrt{6x-5} + \sqrt{3x-2})}$

$= \lim_{x \rightarrow 1^+} \frac{6x-5-3x+2}{(x^2+4x-5)(\sqrt{6x-5} + \sqrt{3x-2})}$

$= \lim_{x \rightarrow 1^+} \frac{3x-3}{(x^2+4x-5)(\sqrt{6x-5} + \sqrt{3x-2})}$

$= \lim_{x \rightarrow 1^+} \frac{3(x-1)}{(x-1)(x+5)(\sqrt{6x-5} + \sqrt{3x-2})} = \frac{3}{6(1+1)} = \frac{1}{2(2)} = \boxed{\frac{1}{4}}$

$\lim_{x \rightarrow 1^-} \left[\frac{|x-1|}{x^2-3x+2} - \frac{3}{4} \right]$

$\lim_{x \rightarrow 1^-} \left[\frac{-(x-1)}{(x-1)(x-2)} - \frac{3}{4} \right] = \frac{-1}{-1} - \frac{3}{4} = \frac{1}{1} - \frac{3}{4} = \boxed{\frac{1}{4}}$

$\therefore f(1) = \lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^-} f(x)$

$\therefore f$ is continuous at $x=1$

Find a nonzero value for the constant A , if any, that makes the function f continuous at $x = 1$, where

$$f(x) = \begin{cases} \frac{A|x-1|}{x^4-1}, & \text{if } x < 1 \\ \frac{1+e^{x-1}}{x}, & \text{if } x \geq 1 \end{cases}$$

Past exam

$$1) f(1) = \frac{1+e^0}{1} = \frac{1+1}{1} = \boxed{2}$$

$$2) \lim_{x \rightarrow 1^+} \frac{1+e^{x-1}}{x} = \frac{1+e^0}{1} = \frac{1+1}{1} = \boxed{2}$$

$$\lim_{x \rightarrow 1^-} \frac{A|x-1|}{x^4-1}$$

$$\lim_{x \rightarrow 1^-} \frac{-A(x-1)}{(x^2-1)(x^2+1)}$$

$$\lim_{x \rightarrow 1^-} \frac{-A(x-1)}{(x-1)(x+1)(x^2+1)} = \frac{-A}{(2)(2)} = \boxed{\frac{-A}{4}}$$

$\therefore f$ is continuous at $x=1$

$$\therefore f(1) = \lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^-} f(x)$$

$$2 = 2 = \frac{-A}{4}$$

$$2 = \frac{-A}{4}$$

$$\Rightarrow \boxed{A = -8}$$

For what value of the number A , that makes the function f continuous at $x = 0$

$$f(x) = \begin{cases} \frac{\frac{1}{(x+2)^2} - \frac{1}{4}}{x}, & \text{if } x \neq 0 \\ 2 - A \sec x, & \text{if } x = 0 \end{cases}$$

Oct 2018

$$1) f(0) = 2 - A \sec 0 = 2 - A \frac{1}{\cos 0} = 2 - A(1) = \boxed{2 - A}$$

$$2) \lim_{x \rightarrow 0} \frac{\frac{1}{(x+2)^2} - \frac{1}{4}}{x} \quad \frac{0}{0}$$

$$= \lim_{x \rightarrow 0} \frac{\frac{4}{4(x+2)^2} - \frac{(x+2)^2}{4(x+2)^2}}{x}$$

$$= \lim_{x \rightarrow 0} \frac{\frac{4 - (x+2)^2}{4(x+2)^2}}{x}$$

$$= \lim_{x \rightarrow 0} \frac{4 - (x+2)^2}{4(x+2)^2 \cdot x}$$

$$= \lim_{x \rightarrow 0} \frac{[2 - (x+2)][2 + (x+2)]}{4(x+2)^2 \cdot x}$$

$$= \lim_{x \rightarrow 0} \frac{[2 - x - 2][4 + x]}{4(x+2)^2 \cdot x}$$

$$= \lim_{x \rightarrow 0} \frac{-x[4 + x]}{4(x+2)^2 \cdot x} = \frac{-4}{4(2)^2} = \boxed{-\frac{1}{4}}$$

$\therefore f$ is continuous at $x=0 \longrightarrow \therefore f(0) = \lim_{x \rightarrow 0} f(x)$

$$*4 \quad 2 - A = -\frac{1}{4}$$

$$2 + \frac{1}{4} = A \longrightarrow A = \frac{8}{4} + \frac{1}{4} \longrightarrow \boxed{A = \frac{9}{4}}$$

For what value of c is f continuous on $(-\infty, \infty)$?

$$f(x) = \begin{cases} cx^2 + 2x & , \quad x < 2 \\ x^3 - cx & , \quad x \geq 2 \end{cases}$$

$$1) \quad f(2) = 8 - 2c$$

$$2) \quad \lim_{x \rightarrow 2^+} (x^3 - cx) = 8 - 2c$$

$$\lim_{x \rightarrow 2^-} (cx^2 + 2x) = 4c + 4$$

$\therefore f$ is continuous on $(-\infty, \infty)$

$\therefore f$ is continuous at $x = 2$

$$\therefore f(2) = \lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^-} f(x)$$

$$8 - 2c = 8 - 2c = 4c + 4$$

$$8 - 2c = 4c + 4$$

$$8 - 4 = 4c + 2c$$

$$6c = 4$$

$$c = \frac{4}{6}$$

$$\therefore c = \frac{2}{3}$$

$$\text{Let } f(x) = \begin{cases} \frac{\sqrt{x-k+1}-1}{x-k}, & \text{if } x > k \\ 2x^2, & \text{if } x \leq k \end{cases}$$

Find all values of k so that f is continuous on $(-\infty, \infty)$

$$f(k) = 2k^2$$

$$\lim_{x \rightarrow k^+} \frac{\sqrt{x-k+1}-1}{x-k} \cdot \frac{\sqrt{x-k+1}+1}{\sqrt{x-k+1}+1}$$

$$\lim_{x \rightarrow k^+} \frac{x-k+1-1}{(x-k)(\sqrt{x-k+1}+1)}$$

$$\lim_{x \rightarrow k^+} \frac{x-k}{(x-k)(\sqrt{x-k+1}+1)} = \frac{1}{\sqrt{1}+1} = \frac{1}{2}$$

$$\lim_{x \rightarrow k^-} 2x^2 = 2k^2$$

$\therefore f$ is continuous on $(-\infty, \infty)$

$\therefore f$ is continuous at $x=k$

$$f(k) = \lim_{x \rightarrow k^+} f(x) = \lim_{x \rightarrow k^-} f(x)$$

$$2k^2 = \frac{1}{2} = 2k^2$$

$$* \frac{1}{2} \quad 2k^2 = \frac{1}{2}$$

$$\frac{1}{2} 2k^2 = \frac{1}{2} \cdot \frac{1}{2} \rightarrow k^2 = \frac{1}{4} \rightarrow k = \pm \frac{1}{2}$$

Find a value for the constant A , if any, that makes the function f continuous at $x = 1$, where

$$f(x) = \begin{cases} \frac{\sqrt{x+1} - \sqrt{2x}}{x-1}, & \text{if } x < 1, x \neq 1 \\ A, & \text{if } x = 1 \end{cases}$$

Mar 2018

$$f(1) = A$$

$$\lim_{x \rightarrow 1^-} \frac{\sqrt{x+1} - \sqrt{2x}}{x-1} \cdot \frac{\sqrt{x+1} + \sqrt{2x}}{\sqrt{x+1} + \sqrt{2x}}$$

$$\lim_{x \rightarrow 1^-} \frac{x+1 - 2x}{(x-1)(\sqrt{x+1} + \sqrt{2x})}$$

$$\lim_{x \rightarrow 1^-} \frac{1 - x}{(x-1)(\sqrt{x+1} + \sqrt{2x})}$$

$$\lim_{x \rightarrow 1^-} \frac{-(x-1)}{(x-1)(\sqrt{x+1} + \sqrt{2x})} = \frac{-1}{\sqrt{2} + \sqrt{2}} = \frac{-1}{2\sqrt{2}}$$

$\therefore f$ is continuous at $x=1$

$$\therefore f(1) = \lim_{x \rightarrow 1^-} f(x)$$

$$A = \frac{-1}{2\sqrt{2}}$$

past exam

$$\text{Let } f(x) = \begin{cases} A + \frac{3|x-1|}{x^2+x-2} & , \text{ if } x < 1, x \neq 2 \\ B & , \text{ if } x = 1 \\ \sqrt{2x-1} & , \text{ if } x > 1 \end{cases}$$

Find the values of A and B so that f is continuous at $x = 1$

$$f(1) = \textcircled{B}$$

$$\lim_{x \rightarrow 1^+} \sqrt{2x-1} = \sqrt{2-1} = \textcircled{1}$$

$$\lim_{x \rightarrow 1^-} \left[A + \frac{3|x-1|}{x^2+x-2} \right]$$

$$\lim_{x \rightarrow 1^-} \left[A + \frac{-3(x-1)}{x^2+x-2} \right]$$

$$\lim_{x \rightarrow 1^-} \left[A + \frac{-3(x-1)}{(x-1)(x+2)} \right]$$

$$= A + \frac{-3}{3} \longrightarrow \textcircled{A-1}$$

$\therefore f$ is continuous at $x=1$

$$f(1) = \lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^-} f(x)$$

$$B = 1 = A - 1$$

$$\boxed{B=1}$$

$$1 = A - 1$$

$$\boxed{A=2}$$

For the values of a and b , if any, that make f continuous everywhere

$$f(x) = \begin{cases} \frac{x^2 - 4}{x - 2}, & \text{if } x < 2 \\ ax^2 - bx + 4, & \text{if } 2 \leq x < 3 \\ 2x - a + b, & \text{if } x \geq 3 \end{cases}$$

Mar 2019

at $x = 2$

$$f(2) = 4a - 2b + 4$$

$$\lim_{x \rightarrow 2^+} (ax^2 - bx + 4) = 4a - 2b + 4$$

$$\lim_{x \rightarrow 2^-} \frac{x^2 - 4}{x - 2} \quad \frac{0}{0}$$

$$\lim_{x \rightarrow 2^-} \frac{(x-2)(x+2)}{x-2} = 4$$

$\therefore f$ is continuous everywhere

$\therefore f$ is continuous at $x = 2$

$$\therefore f(2) = \lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^-} f(x)$$

$$4a - 2b + 4 = 4a - 2b + 4 = 4$$

$$4a - 2b + 4 = 4$$

$$\therefore 4a - 2b \xrightarrow{\div 2} b = 2a \rightarrow (1)$$

(1) in (2) $5a - 4a - 1 = 0$

$$a = 1 \xrightarrow{\text{in (1)}} b = 2$$

at $x = 3$

$$f(3) = 6 - a + b$$

$$\lim_{x \rightarrow 3^+} (2x - a + b) = 6 - a + b$$

$$\lim_{x \rightarrow 3^-} (ax^2 - bx + 4) = 9a - 3b + 4$$

$\therefore f$ is continuous everywhere

$\therefore f$ is continuous at $x = 3$

$$\therefore f(3) = \lim_{x \rightarrow 3^+} f(x) = \lim_{x \rightarrow 3^-} f(x)$$

$$6 - a + b = 6 - a + b = 9a - 3b + 4$$

$$6 - a + b = 9a - 3b + 4$$

$$\div 2 \quad 10a - 4b - 2 = 0$$

$$5a - 2b - 1 = 0 \rightarrow (2)$$



Text book question



Suppose that f and g are continuous function such that $g(2) = 6$ and $\lim_{x \rightarrow 2} [3f(x) + f(x)g(x)] = 36$. Find $f(2)$.

$\therefore f$ and g are continuous $\begin{cases} \therefore f(2) = \lim_{x \rightarrow 2} f(x) \\ \therefore g(2) = \lim_{x \rightarrow 2} g(x) \end{cases}$

$$\lim_{x \rightarrow 2} (3f(x) + f(x)g(x)) = 36$$

$$3f(2) + f(2)g(2) = 36$$

$$3f(2) + f(2) \cdot 6 = 36$$

$$9f(2) = 36$$

$$f(2) = \frac{36}{9}$$

$$\therefore f(2) = 4$$



Discontinuity



خطوات الحل

$$f(a)$$

(1) أوجد الصورة . ←

$$\lim_{x \rightarrow a} f(x)$$

(2) أوجد اللمت . ←

(3) تحديد نوع عدم الاتصال

إذا كان ناتج اللمت $= \infty$ "حتى لو كان أحد الأطراف دون الآخر"

$$\lim_{x \rightarrow a} f(x) = \infty$$

infinite discontinuity

إذا كان ناتج اللمت بالاتجاهين مختلف

$$\lim_{x \rightarrow a^+} f(x) \neq \lim_{x \rightarrow a^-} f(x)$$

jump discontinuity

إذا كانت الصورة $\neq$ اللمت

$$f(a) \neq \lim_{x \rightarrow a} f(x)$$

removable discontinuity

إذا لم يحدد نقاط .. يتم دراسة

أصفار المقام بشرط انتمائها للمجال

المفاصل



Text book questions



Find the numbers at which f is discontinuous, then classify the type of discontinuity point (if any)

$$f(x) = \begin{cases} x^2, & \text{if } x < -1 \\ x, & \text{if } -1 \leq x < 1 \\ \frac{1}{x}, & \text{if } x \geq 1 \end{cases}$$

~~$x \neq 0$~~

at $x = -1$

$$f(-1) = -1$$

$$\lim_{x \rightarrow -1^+} x = -1$$

$$\lim_{x \rightarrow -1^-} x^2 = (-1)^2 = 1$$

$$\therefore \lim_{x \rightarrow -1^+} f(x) \neq \lim_{x \rightarrow -1^-} f(x)$$

$\therefore$ at $x = -1 \longrightarrow$ Jump discontinuity.

at $x = 1$

$$f(1) = \frac{1}{1} = 1$$

$$\lim_{x \rightarrow 1^+} \frac{1}{x} = \frac{1}{1} = 1$$

$$\lim_{x \rightarrow 1^-} x = 1 = 1$$

$$\therefore f(1) = \lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^-} f(x)$$

$\therefore$ at $x = 1 \longrightarrow f(x)$ is continuous

$$\text{Let } f(x) = \begin{cases} \frac{6x-6}{|x-1|} & , \text{ if } x < 1 \\ \frac{x^3-1}{\sqrt{x}-1} & , \text{ if } x > 1 \end{cases}$$

Classify the discontinuity of $f(x)$

at $x=1$

$f(1)$ undefined

$$\lim_{x \rightarrow 1^+} \frac{x^3-1}{\sqrt{x}-1} \quad \frac{0}{0} \quad * \quad \frac{\sqrt{x}+1}{\sqrt{x}+1}$$

$$\lim_{x \rightarrow 1^+} \frac{(x^3-1)(\sqrt{x}+1)}{x-1}$$

$$\lim_{x \rightarrow 1^+} \frac{(x-1)(x^2+x+1)(\sqrt{x}+1)}{x-1} = 3(2) = 6$$

$$\lim_{x \rightarrow 1^-} \frac{6x-6}{|x-1|}$$

$$\lim_{x \rightarrow 1^-} \frac{6(x-1)}{-(x-1)} = \frac{6}{-1} = -6$$

$$\therefore \lim_{x \rightarrow 1^+} f(x) \neq \lim_{x \rightarrow 1^-} f(x)$$

$\Rightarrow$ at $x=1 \Rightarrow$ Jump discontinuity

$$\text{Let } f(x) = \begin{cases} \frac{2x+1}{x+3}, & \text{if } x < 0 \\ \frac{x^2-1}{x^2+x-2}, & \text{if } x \geq 0 \end{cases}$$

$\rightarrow -3 \checkmark$
 $(x-1)(x+2)=0$
 $x=1 \checkmark$ $x=-2 \times$

Classify the discontinuity of $f(x)$ as removable, jump or infinite

at $x=0$

0
-3 1

$$f(0) = \frac{0-1}{0+0-2} = \frac{-1}{-2} = \frac{1}{2}$$

$$\lim_{x \rightarrow 0^+} \frac{x^2-1}{x^2+x-2} = \frac{0-1}{0+0-2} = \frac{1}{2}$$

$$\lim_{x \rightarrow 0^-} \frac{2x+1}{x+3} = \frac{0+1}{0+3} = \frac{1}{3}$$

$$\therefore \lim_{x \rightarrow 0^+} f(x) \neq \lim_{x \rightarrow 0^-} f(x)$$

$\therefore$ at $x=0 \Rightarrow$ Jump discontinuity

at $x=-3$

$f(-3)$ undefined

$$\lim_{x \rightarrow -3} \frac{2x+1}{x+3} = \frac{-6+1}{0} = \frac{-5}{0} = \infty$$

$$\lim_{x \rightarrow -3^+} \frac{2x+1}{x+3} = \frac{-5}{0^+} = -\infty \quad \Bigg| \quad \lim_{x \rightarrow -3^-} \frac{2x+1}{x+3} = \frac{-5}{0^-} = +\infty$$

$\Rightarrow \therefore$ at $x=-3 \Rightarrow$ infinite discontinuity

at $x=1$

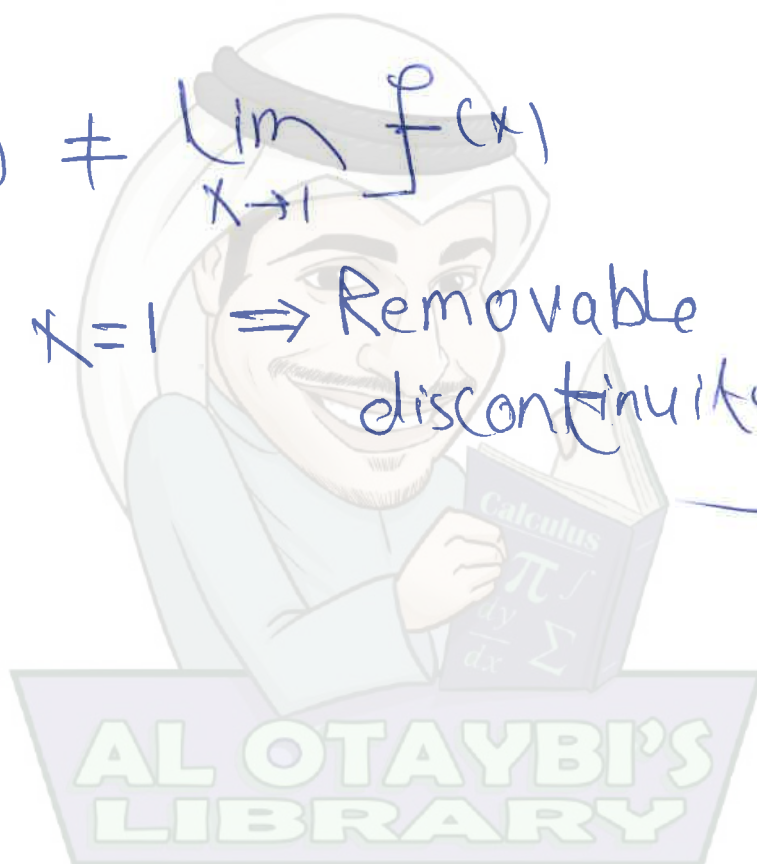
$f(1)$ undefined

$$\lim_{x \rightarrow 1} \frac{x^2 - 1}{x^2 + x - 2} \quad \frac{0}{0}$$

$$\lim_{x \rightarrow 1} \frac{\cancel{(x-1)}(x+1)}{\cancel{(x-1)}(x+2)} = \left(\frac{2}{3}\right)$$

$$\therefore f(1) \neq \lim_{x \rightarrow 1} f(x)$$

$\therefore$ at $x=1 \Rightarrow$ Removable discontinuity



$$f(x) = \begin{cases} \frac{x^2 + x - 6}{x + 3}, & \text{if } x < 0 \\ \frac{x + 1}{(x - 1)^2}, & \text{if } x \geq 0 \end{cases}$$

Nov 2016

-3 $\neq$ 0
-3 $\neq$ 0

Find all the points of discontinuity of f .

1 $\neq$ 0
1 $\neq$ 0

Classify each discontinuity as removable, jump, or infinite

at $x = 0$

$$f(0) = \frac{0+1}{(0-1)^2} = \frac{1}{1} = 1$$

$$\lim_{x \rightarrow 0^+} \frac{x+1}{(x-1)^2} = \frac{1}{1} = 1$$

$$\lim_{x \rightarrow 0^-} \frac{x^2+x-6}{x+3} = \frac{0+0-6}{0+3} = -2$$

$$\because \lim_{x \rightarrow 0^+} f(x) \neq \lim_{x \rightarrow 0^-} f(x)$$

$\therefore$ at $x = 0 \rightarrow$ Jump discontinuity.

at $x = -3$

$f(-3)$ undefined

$$\lim_{x \rightarrow -3} \frac{x^2+x-6}{x+3} \quad \frac{0}{0}$$

$$\lim_{x \rightarrow -3} \frac{(x-2)(x+3)}{x+3} = -5$$

at $x = -3 \rightarrow$ Removable discontinuity

at $x = 1$

$f(1)$ undefined

$$\lim_{x \rightarrow 1} \frac{x+1}{(x-1)^2} = \frac{2}{0} = \infty$$

$$\lim_{x \rightarrow 1^+} \frac{x+1}{(x-1)^2} = \frac{2}{0^+} = +\infty$$

$$\lim_{x \rightarrow 1^-} \frac{x+1}{(x-1)^2} = \frac{2}{0^+} = +\infty$$

$\therefore$ at $x = 1 \rightarrow$ infinite discontinuity.



a) Classify the points of discontinuity of f .

b) On what interval(s) is the function f continuous.

$$f(x) = \begin{cases} \frac{x^2 - x}{x^3}, & x < 0 \\ \frac{x - 1}{x^2 - x}, & 0 < x < 1 \\ \frac{\sin \pi x}{\pi x}, & x > 1 \end{cases}$$

Annotations:

- For $\frac{x^2 - x}{x^3}$, $x < 0$: $x = 0$
- For $\frac{x - 1}{x^2 - x}$, $0 < x < 1$: $x^2 - x = 0$
 $x(x - 1) = 0$
 $x = 0$ $x = 1$
- For $\frac{\sin \pi x}{\pi x}$, $x > 1$: $x = 0$

at $x = 0$

$f(0)$ undefined

$$\lim_{x \rightarrow 0^+} \frac{x - 1}{x^2 - x}$$

$$\lim_{x \rightarrow 0^+} \frac{\cancel{x} - 1}{x(\cancel{x} - 1)} = \frac{1}{0^+} = +\infty$$

$$\lim_{x \rightarrow 0^-} \frac{x^2 - x}{x^3}$$

$$\lim_{x \rightarrow 0^-} \frac{\cancel{x}(\cancel{x} - 1)}{\cancel{x}^3} = \frac{-1}{0^+} = -\infty$$

$\therefore$ at $x = 0 \longrightarrow$ infinite discontinuity.

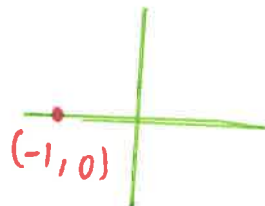
at $x=1$

$f(x)$ undefined

$$\lim_{x \rightarrow 1^+} \frac{\sin \pi x}{\pi x}$$

~~$\neq 1$~~

$$= \frac{\sin \pi}{\pi} = \frac{0}{\pi} = 0$$



$$\lim_{x \rightarrow 1^-} \frac{x-1}{x^2-x} = \frac{0}{0}$$

$$\lim_{x \rightarrow 1^-} \frac{x-1}{x(x-1)} = \frac{1}{1} = 1$$

$$\therefore \lim_{x \rightarrow 1^+} f(x) \neq \lim_{x \rightarrow 1^-} f(x)$$

$\therefore$ at $x=1 \rightarrow$ Jump discontinuity

b) $f(x)$ is continuous on

$$\mathbb{R} \setminus [0, 1]$$

or

$$= (-\infty, 0) \cup (0, 1) \cup (1, \infty)$$

نقاط
عدم
الاتصال

$$\text{Let } f(x) = \begin{cases} x^2 - \ln(x + e), & \text{if } -1 < x \leq 0 \\ e^x - \sin x, & \text{if } x > 0 \end{cases}$$

a) Find and classify the discontinuities of f as removable, infinite, or jump.

b) Does $\lim_{x \rightarrow 0} (f(x))^2$ exist? explain why or why not?

Oct 2018

at $x=0$

$$f(0) = 0 - \ln(0 + e) = -\ln e = \boxed{-1}$$

$$\lim_{x \rightarrow 0^+} (e^x - \sin x) = e^0 - 0 = \boxed{1}$$

$$\lim_{x \rightarrow 0^-} (x^2 - \ln(x + e)) = 0 - \ln e = \boxed{-1}$$

$$\therefore \lim_{x \rightarrow 0^+} f(x) \neq \lim_{x \rightarrow 0^-} f(x)$$

$\therefore$ at $x=0 \longrightarrow$ Jump discontinuity.

$$\text{b) } \lim_{x \rightarrow 0} (f(x))^2$$

$$\lim_{x \rightarrow 0^+} (f(x))^2 = (1)^2 = 1$$

$$\lim_{x \rightarrow 0^-} (f(x))^2 = (-1)^2 = 1$$

$$\implies \therefore \lim_{x \rightarrow 0} (f(x))^2 = 1$$

Define $f(x) = \tan^{-1}\left(\frac{1}{x-2}\right)$ for $x \neq 2$

Mat 2018

a) Explain why f is discontinuous at $x = 2$

b) Classify the type of this discontinuity

a) $f(2) = \tan^{-1}\left(\frac{1}{0}\right)$ undefined

$\Rightarrow \therefore f$ is Not continuous at $x=2$.

b) $\lim_{x \rightarrow 2} \tan^{-1}\left(\frac{1}{x-2}\right) = \tan^{-1}\left(\frac{1}{0}\right) = \tan^{-1}(\infty)$

$$\lim_{x \rightarrow 2^+} \tan^{-1}\left(\frac{1}{x-2}\right) = \tan^{-1}\left(\frac{1}{0}\right) = \tan^{-1}(+\infty) = \frac{\pi}{2} \quad \Bigg| \quad \lim_{x \rightarrow 2^-} \tan^{-1}\left(\frac{1}{x-2}\right) = \tan^{-1}\left(\frac{1}{0}\right) = \tan^{-1}(-\infty) = -\frac{\pi}{2}$$

$$\therefore \lim_{x \rightarrow 2^+} f(x) \neq \lim_{x \rightarrow 2^-} f(x)$$

$\therefore$ at $x=2$ there is Jump discontinuity.

a) Find the points of discontinuity of f and classify each discontinuity as jump, infinite, or removable.

b) Find the interval of continuity of f

$$f(x) = \frac{|x-1|(x-2)}{(x^2-3x+2)(x-4)}$$

$$(x^2-3x+2)(x-4) = 0$$

$$(x-1)(x-2)(x-4) = 0$$

$$x=1$$

$$x=2$$

$$x=4$$

at $x=1$

$f(1)$ undefined

$$\lim_{x \rightarrow 1} \frac{|x-1|(x-2)}{(x^2-3x+2)(x-4)}$$

$$\lim_{x \rightarrow 1^+} \frac{(x-1)(x-2)}{(x^2-3x+2)(x-4)}$$

$$\lim_{x \rightarrow 1^+} \frac{\cancel{(x-1)}(\cancel{x-2})}{\cancel{(x-1)}(\cancel{x-2})(x-4)}$$

$$= \frac{1}{-3} = -\frac{1}{3}$$

$$\lim_{x \rightarrow 1^-} \frac{-(x-1)(x-2)}{(x^2-3x+2)(x-4)}$$

$$\lim_{x \rightarrow 1^-} \frac{-\cancel{(x-1)}(\cancel{x-2})}{\cancel{(x-1)}(\cancel{x-2})(x-4)}$$

$$= \frac{-1}{-3} = \frac{1}{3}$$

$$\therefore \lim_{x \rightarrow 1^+} f(x) \neq \lim_{x \rightarrow 1^-} f(x)$$

$\therefore$ at $x=1 \longrightarrow$ Jump discontinuity.

at $x=2$

$f(2)$ unde find

$$\lim_{x \rightarrow 2} \frac{|x-1| (x-2)}{(x^2-3x+2) (x-4)}$$

$$\lim_{x \rightarrow 2} \frac{+ (x-1) (x-2)}{(x-1) (x-2) (x-4)} = \frac{1}{-2} = -\frac{1}{2}$$

$$\therefore f(2) \neq \lim_{x \rightarrow 2} f(x)$$

$\therefore$ at $x=2 \longrightarrow$ removable discontinuity.

at $x=4$

$f(4)$ unde find

$$\lim_{x \rightarrow 4} \frac{|x-1| (x-2)}{(x^2-3x+2) (x-4)}$$

$$\lim_{x \rightarrow 4} \frac{+ (x-1) (x-2)}{(x-1) (x-2) (x-4)} = \frac{1}{0} = \infty$$

$$\lim_{x \rightarrow 4^+} \frac{1}{x-4} = \frac{1}{0} = +\infty \quad \parallel \quad \lim_{x \rightarrow 4^-} \frac{1}{x-4} = \frac{1}{0} = -\infty$$

$\therefore$ at $x=4 \longrightarrow$ infinite discontinuity.

b) The interval of continuity of f

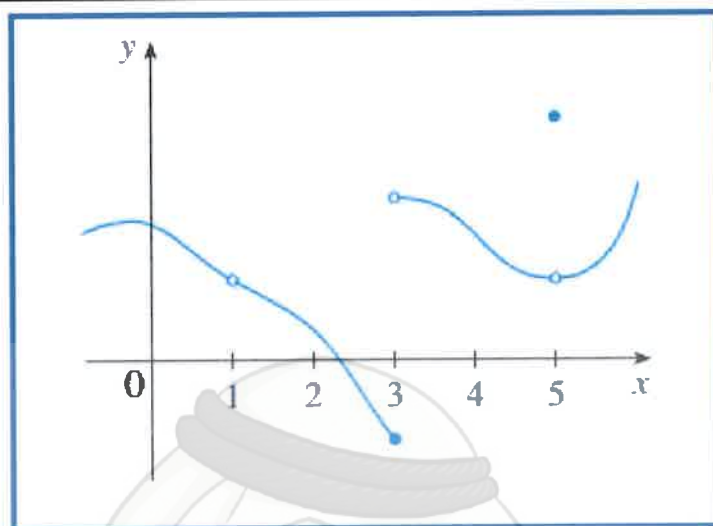
$$= \mathbb{R} \setminus \{1, 2, 4\} = (-\infty, 1) \cup (1, 2) \cup (2, 4) \cup (4, \infty)$$



Text book question



Find the x-coordinate of discontinuity point, and classify them (do NOT justify). Then find the interval of continuity



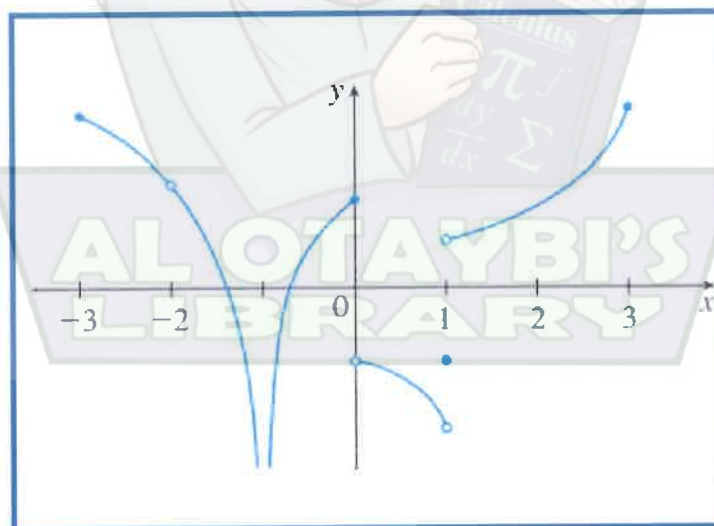
at $x=1 \rightarrow$ removable discontinuity.

at $x=3 \rightarrow$ Jump discontinuity.

at $x=5 \rightarrow$ removable discontinuity.

نقطة الالتقاء: 3

نقطة الفجوة: 1, 5



at $x=-2 \rightarrow$ removable discontinuity

at $x=-1 \rightarrow$ infinite discontinuity.

at $x=0 \rightarrow$ Jump discontinuity.

at $x=1 \rightarrow$ Jump discontinuity.

نقطة الالتقاء: -1, 0, 1

نقطة الفجوة: -2

Q1. [10+10=20 pts.] Use the graph of $y = f(x)$ shown below to:

- a) Find each of the following limits if it exists, or explain why if it does not exist:
 i) $\lim_{x \rightarrow -4^+} f(x)$; ii) $\lim_{x \rightarrow -4^-} f(x)$; iii) $\lim_{x \rightarrow 2^-} f(x)$; iv) $\lim_{x \rightarrow 2^+} f(x)$; v) $\lim_{x \rightarrow 4} f(x)$.
 b) Find and classify the discontinuities of f as removable, infinite or jump.

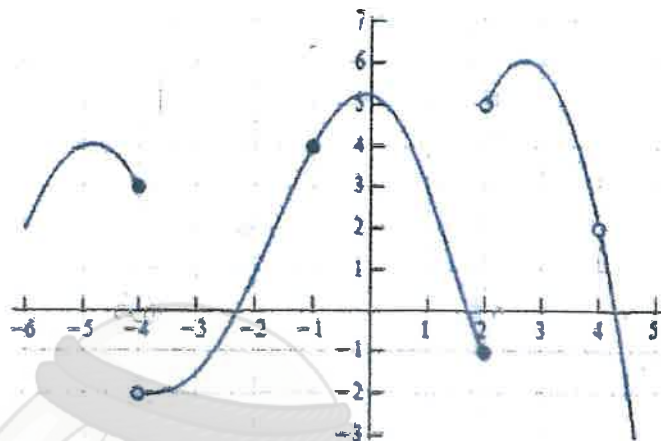


Figure 1: The graph of $y = f(x)$.

i) $\lim_{x \rightarrow -4^+} f(x) = -2$

ii) $\lim_{x \rightarrow -4^-} f(x) = 3$

iii) $\lim_{x \rightarrow 2^-} f(x) = -1$

iv) $\lim_{x \rightarrow 2^+} f(x) = 5$

v) $\lim_{x \rightarrow 4} f(x) = 2$

Extra

$f(-4) = 3$

$f(2) = -1$

$f(4) = \text{undefined}$

يفرق الـ 4 : 2
يفرق الـ 4 : 4

b)

at $x=4 \rightarrow$ removable discontinuity
 $(f(4) \neq \lim_{x \rightarrow 4} f(x))$
 undefined $\neq 2$

at $x=2 \rightarrow$ Jump discontinuity
 $(\lim_{x \rightarrow 2^+} f(x) \neq \lim_{x \rightarrow 2^-} f(x))$
 $5 \neq -1$

at $x=-4 \rightarrow$ Jump discontinuity
 $(\lim_{x \rightarrow -4^+} f(x) \neq \lim_{x \rightarrow -4^-} f(x))$
 $-2 \neq 3$

[14+6=20 pts.] Use the graph of $y = f(x)$ shown below to:

a) Find the following limits:

i) $\lim_{x \rightarrow -8^+} f(x)$; ii) $\lim_{x \rightarrow -8^-} f(x)$; iii) $\lim_{x \rightarrow -2^+} f(x)$; iv) $\lim_{x \rightarrow -2^-} f(x)$; v) $\lim_{x \rightarrow 6^+} f(x)$; vi) $\lim_{x \rightarrow 6^-} f(x)$; vii) $\lim_{x \rightarrow 2} f(x)$.

b) Find and classify the discontinuities of f as removable, infinite or jump.

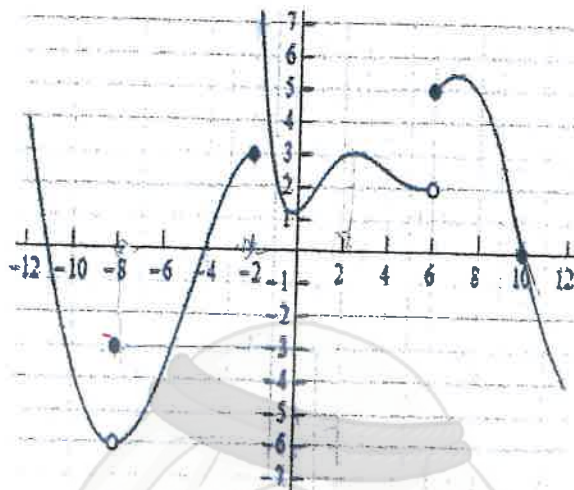


Figure 1: The graph of $y = f(x)$.

i) $\lim_{x \rightarrow -8^+} f(x) = -6$

ii) $\lim_{x \rightarrow -8^-} f(x) = -6$

iii) $\lim_{x \rightarrow -2^+} f(x) = \infty$

iv) $\lim_{x \rightarrow -2^-} f(x) = 3$

v) $\lim_{x \rightarrow 6^+} f(x) = 5$

vi) $\lim_{x \rightarrow 6^-} f(x) = 2$

vii) $\lim_{x \rightarrow 2} f(x) = 3$

يفرق الـ ٢ : -2
يفرق الـ ٨ : -8

Extra

$f(-8) = -3$

$f(-2) = 3$

$f(6) = 5$

Equation of vertical Asym.

$x = -2$ ($\lim_{x \rightarrow -2^+} f(x) = \infty$)

$\lim_{x \rightarrow -2^-} f(x) = 3$

b)

at $x = -8 \rightarrow$ removable discontinuity

($f(-8) \neq \lim_{x \rightarrow -8} f(x)$)

$-3 \neq -6$

at $x = 6 \rightarrow$ jump discontinuity

($\lim_{x \rightarrow 6^+} f(x) \neq \lim_{x \rightarrow 6^-} f(x)$)

$5 \neq 2$

at $x = -2 \rightarrow$ infinite discontinuity

($\lim_{x \rightarrow -2^+} f(x) = \infty$, $\lim_{x \rightarrow -2^-} f(x) = 3$)

[15+5=20 pts.] Use the graphs of $y = f(x)$ and $y = g(x)$ shown below to:

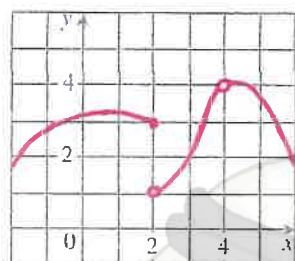
past exam

a) Find each of the following limits (if it exists):

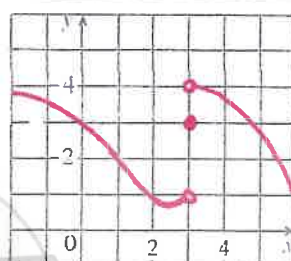
i) $\lim_{x \rightarrow 4} f(x)$; ii) $\lim_{x \rightarrow 3} g(x)$; iii) $\lim_{x \rightarrow 3^+} f(x)g(x)$; vi) $\lim_{x \rightarrow 3^-} f(x)g(x)$; v) $\lim_{x \rightarrow 3} f(x)g(x)$.

b) Find and classify the discontinuities of f and g as removable, infinite or jump.

يفرق الاتجاهين: 2
يفرق المبررة: 4



The graph of $y = f(x)$



The graph of $y = g(x)$

يفرق الاتجاهين: 3

يفرق المبررة:

i) $\lim_{x \rightarrow 4} f(x) = 4$

ii) $\lim_{x \rightarrow 3} g(x) = \text{DNE}$ ($\lim_{x \rightarrow 3^+} g(x) \neq \lim_{x \rightarrow 3^-} g(x)$)
 $4 \neq 1$

iii) $\lim_{x \rightarrow 3^+} f(x)g(x) = 2(4) = 8$

iv) $\lim_{x \rightarrow 3^-} f(x)g(x) = 2(1) = 2$

v) $\lim_{x \rightarrow 3} f(x)g(x) = \text{DNE}$ ($\lim_{x \rightarrow 3^+} f(x)g(x) \neq \lim_{x \rightarrow 3^-} f(x)g(x)$)
 $8 \neq 2$

b) f

at $x = 4 \rightarrow$ removable discontinuity

($f(4) \neq \lim_{x \rightarrow 4} f(x)$)
undefined $\neq 4$

at $x = 2 \rightarrow$ Jump discontinuity

($\lim_{x \rightarrow 2^+} f(x) \neq \lim_{x \rightarrow 2^-} f(x)$)
 $1 \neq 3$

g

at $x = 3 \rightarrow$ jump discontinuity

($\lim_{x \rightarrow 3^+} g(x) \neq \lim_{x \rightarrow 3^-} g(x)$)
 $4 \neq 1$



Intermediate Value Theorem (I.V.T)



التعريف



Suppose that f is continuous on the closed interval $[a, b]$ and let N be any real number between $f(a)$ and $f(b)$, where $f(a) \neq f(b)$. Then there exists a number c in (a, b) such that $f(c) = N$

نظرية تستخدم لإثبات أن للدالة حل حقيقي واحد (على الأقل) ..
ولا تستخدم لإيجاد هذا الحل



خطوات الحل



- 1) التعويض بعددين أحدهما موجب والآخر سالب (مالم يذكر غير ذلك).
- 2) اثبات اتصال الدالة بفترة أعداد التعويض.
- 3) كتابة التعريف.



انتبه



التعريف (لا!!!!!! ازم) تعرفه , ومن الممكن أن تسأل عنه كسؤال منفصل.

Prove that the equation has at least one solution

$$x^5 - 3x^4 - 2x^3 - x + 1 = 0$$

Let $f(x) = x^5 - 3x^4 - 2x^3 - x + 1$

1) $f(0) = 0 - 0 - 0 - 0 + 1 = 1 > 0$

2) $f(1) = 1 - 3 - 2 - 1 + 1 = -4 < 0$

From ①② $\longrightarrow$ By I.V.T

Since $f(x)$ is continuous on $\mathbb{R}$ [polynomial of degree 5]

then $f(x)$ is continuous on $[0, 1]$

then there exist at least one $c \in (0, 1)$

such that $f(c) = 0$

بما أن الدالة متصلة في مجالها
إذاً الدالة متصلة في [الفترة المغلقة]
إذاً يوجد حل واحد على الأقل في (الفترة المفتوحة)





Text book question



Use the Intermediate Value Theorem to show that there is a root of the given equation in the specified interval.

$$x^4 + x - 3 = 0 \quad , \quad (1, 2)$$

Let $f(x) = x^4 + x - 3$

1) $f(1) = 1 + 1 - 3 = -1 < 0$

2) $f(2) = 2^4 + 2 - 3$
 $= 16 - 1 = 15 > 0$

From ① ② $\longrightarrow$ By I.V.T

Since $f(x)$ is continuous on $\mathbb{R}$ [polynomial of degree 4]

then $f(x)$ is continuous on $[1, 2]$

then there exist at least one $c \in (1, 2)$

such that $f(c) = 0$

Use Intermediate Value Theorem to show that the equation

$$x^2 \cos x + 3x - 1 = 0 \text{ has a real root}$$

Let $f(x) = x^2 \cos x + 3x - 1$

1) $f(0) = 0 + 0 - 1 = -1 < 0$

2) $f\left(\frac{\pi}{2}\right) = \left(\frac{\pi}{2}\right)^2 \cos \frac{\pi}{2} + 3\frac{\pi}{2} - 1$

$= 3\frac{\pi}{2} - 1 > 0$

From ①② $\longrightarrow$ By I.V.T

Since $f(x)$ is continuous on $\mathbb{R}$ [$\cos x$: Trigonometric fun.]
 $x^2, 3x$: polynomials]

then $f(x)$ is continuous on $\left[0, \frac{\pi}{2}\right]$

then there exist at least one $c \in \left(0, \frac{\pi}{2}\right)$

such that $f(c) = 0$

Use Intermediate Value Theorem to show that the equation

$$\sin\left(\frac{\pi x}{2}\right) - \frac{3}{4} = 0 \quad \text{has a real root}$$

Let $f(x) = \sin\left(\frac{\pi x}{2}\right) - \frac{3}{4}$

1) $f(0) = \sin 0 - \frac{3}{4} = -\frac{3}{4} < 0$

2) $f(1) = \sin\left(\frac{\pi}{2}\right) - \frac{3}{4}$
 $= 1 - \frac{3}{4} = \frac{4}{4} - \frac{3}{4} = \frac{1}{4} > 0$

From ①② $\longrightarrow$ By I.V.T

Since $f(x)$ is continuous on $\mathbb{R}$ [Trigonometric Function]

then $f(x)$ is continuous on $[0, 1]$

then there exist at least one $c \in (0, 1)$

such that $f(c) = 0$

Show that the equation

$3\sin x + 2x + 1 = 0$ has a real root.

Let $f(x) = 3\sin x + 2x + 1$

1) $f(0) = 0 + 0 + 1 = 1 > 0$

2) $f(-\pi) = 3\sin(-\pi) - 2\pi + 1$
 $= -3\sin(\pi) - 2\pi + 1 = -2\pi + 1 < 0$

From ①② $\longrightarrow$ By I.V.T

Since $f(x)$ is continuous on $\mathbb{R}$ [$\sin x$: Trigonometric function
 $2x+1$: polynomial]

then $f(x)$ is continuous on $[-\pi, 0]$

then there exist at least one $c \in (-\pi, 0)$

such that $f(c) = 0$



Text book question



Prove that the equation has at least one real root

$$100e^{-x/100} = 0.01x^2$$

$$100e^{-\frac{x}{100}} - 0.01x^2 = 0$$

$$\text{Let } f(x) = 100e^{-\frac{x}{100}} - \frac{1}{100}x^2$$

$$1) f(0) = 100e^0 - 0 = 100 > 0$$

$$\begin{aligned} 2) f(100) &= 100e^{-\frac{100}{100}} - \frac{1}{100}(100)^2 \\ &= 100e^{-1} - 100 \\ &= 100(e^{-1} - 1) = 100\left(\frac{1}{e} - 1\right) < 0 \end{aligned}$$

From ① ② $\longrightarrow$ By I.V.T

Since $f(x)$ is continuous on $\mathbb{R}$ $\left[\begin{array}{l} e^{-\frac{x}{100}} : \text{exponential fun.} \\ x^2 : \text{polynomial} \end{array} \right]$
then $f(x)$ is continuous on $[0, 100]$

then there exist at least one $c \in (0, 100)$

such that $f(c) = 0$



Text book question



Prove that the equation has at least one real root

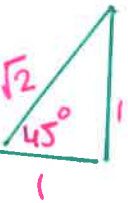
$$\arctan x = 1 - x$$

$$\tan^{-1}x + x - 1 = 0$$

$$\text{Let } f(x) = \tan^{-1}x + x - 1$$

$$1) f(0) = 0 + 0 - 1 = -1 < 0$$

$$2) f(1) = \tan^{-1}1 + 1 - 1 = \tan^{-1}1 = \frac{\pi}{4} > 0$$



From ① ② $\longrightarrow$ By I.V.T

Since $f(x)$ is continuous on $\mathbb{R}$ [$\tan^{-1}x$: inverse of Trigonometric function]

then $f(x)$ is continuous on $[0, 1]$ [x : polynomial]

then there exist at least one $c \in (0, 1)$

such that $f(c) = 0$

Use the Intermediate Value Theorem to show that the equation

$$e^x + \cos x = 4 \text{ has a real root}$$

July 2017

$$e^x + \cos x - 4 = 0$$

$$\text{Let } f(x) = e^x + \cos x - 4$$

$$1) f(0) = e^0 + \cos 0 - 4 = 1 + 1 - 4 = -2 < 0$$

$$2) f(2\pi) = e^{2\pi} + \cos 2\pi - 4 = e^{2\pi} + 1 - 4 = e^{2\pi} - 3 > 0$$

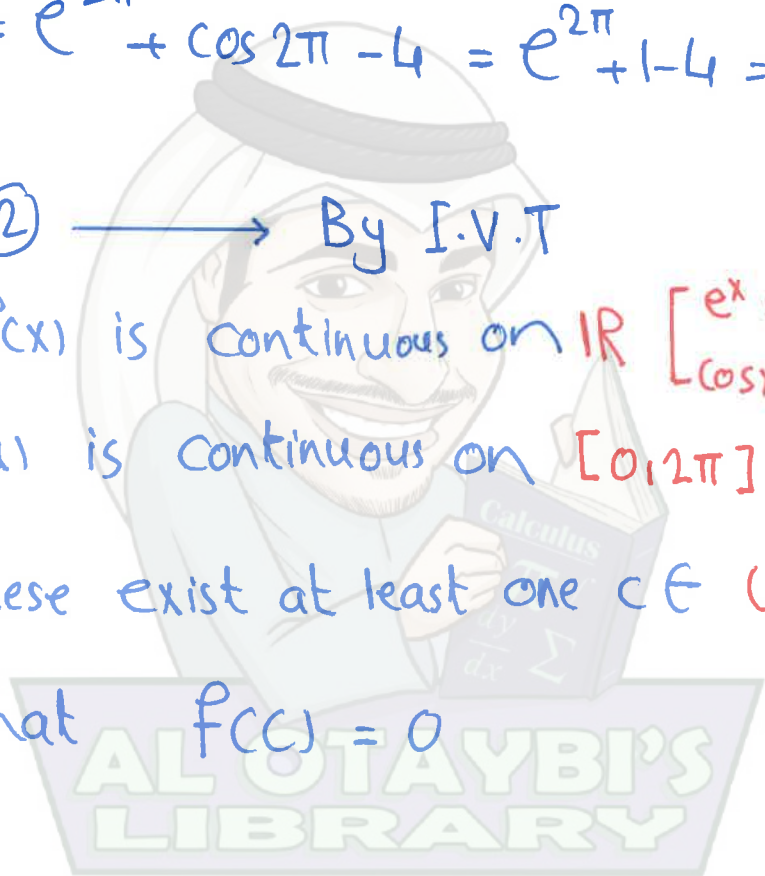
From ① ② $\longrightarrow$ By I.V.T

since $f(x)$ is continuous on $\mathbb{R}$ [e^x : exponential function
 $\cos x$: Trigonometric function]

then $f(x)$ is continuous on $[0, 2\pi]$

then there exist at least one $c \in (0, 2\pi)$

such that $f(c) = 0$



Let $f(x) = x - \ln(x^3)$

Use the intermediate value theorem to show that $f(x)$ has at least one real root on $[1, e]$

$$\begin{aligned} \text{Let } f(x) &= x - \ln x^3 \\ &= x - 3 \ln x \end{aligned}$$

$$\textcircled{1} \text{ Let } f(1) = 1 - 3 \ln 1 = 1 - 0 = 1 > 0$$

$$\begin{aligned} \textcircled{2} \text{ Let } f(e) &= e - 3 \ln e = e - 3(1) \\ &= e - 3 \approx 2.7 - 3 < 0 \end{aligned}$$

From $\textcircled{1} \textcircled{2} \rightarrow$ By I.V.T

since $f(x)$ is continuous on $(0, \infty)$ [logarithmic function]

then $f(x)$ is continuous on $[1, e]$

then there exist at least one $c \in (1, e)$

such that $f(c) = 0$



Text book question



Use the Intermediate Value Theorem to show that there is a root of the given equation in the specified interval.

$$\sin x = x^2 - x, \quad (1, 2)$$

$$\sin x - x^2 + x = 0$$

Let $f(x) = \sin x - x^2 + x$

$$1) \quad f(1) = \sin 1 - 1 + 1 = \sin 1 > 0$$

$$2) \quad f(2) = \sin 2 - 4 + 2 = \sin 2 - 2 < 0 \quad \begin{array}{l} \text{since} \\ \sin 2 < 1 \end{array}$$

From ① ② $\longrightarrow$ By I.V.T

Since $f(x)$ is continuous on $\mathbb{R}$ $\left[\begin{array}{l} \sin x: \text{Trigonometric fun} \\ x^2, x: \text{polynomials} \end{array} \right]$
then $f(x)$ is continuous on $[1, 2]$

then there exist at least one $c \in (1, 2)$

such that $f(c) = 0$

a) State the Intermediate Value Theorem.

b) Show that there is at least one real root of the equation

$$\cos(\sqrt{x}) = e^x - 2$$

Mar 2019

a) Suppose that f is continuous on the closed interval $[a, b]$ and let N be any real number between $f(a)$ and $f(b)$, where $f(a) \neq f(b)$.

Then there exist a number c is (a, b) such that $f(c) = N$

b) $\cos(\sqrt{x}) - e^x + 2 = 0$

Let $f(x) = \cos(\sqrt{x}) - e^x + 2$

1) $f(0) = \cos 0 - e^0 + 2 = 1 - 1 + 2 = 2 > 0$

2) $f(2) = \cos \sqrt{2} - e^2 + 2 < 0$

Since $-\frac{1}{\sqrt{e}} < \cos \sqrt{2} < \frac{1}{\sqrt{e}}$

$$e^2 \approx (2.7)^2$$

From ① ② $\longrightarrow$ By I.V.T

Since $f(x)$ is continuous on $[0, \infty)$ [$\sqrt{x}$ root function]

then $f(x)$ is continuous on $[0, 2]$

then there exist at least one $c \in (0, 2)$

such that $f(c) = 0$

Show that the equation

$$e^x = 3 - 2x$$

Has a real solution on $[0, 1]$

$$e^x - 3 + 2x = 0$$

Let $f(x) = e^x - 3 + 2x$

① $f(0) = e^0 - 3 + 2(0) = 1 - 3 + 0 = -2 < 0$

② $f(1) = e^1 - 3 + 2(1) = e - 1 \approx 2.71 - 1 > 0$

From ① ② $\rightarrow$ By I.V.T

Since $f(x)$ is continuous on $\mathbb{R}$ [exponential and polynomial]

then $f(x)$ is continuous on $[0, 1]$

then there exist at least one $c \in (0, 1)$

such that $f(c) = 0$

Show that the graph f and g are intersect where

$$f(x) = x^3 + 7x^2 - 5 \quad \text{and} \quad g(x) = 5 - x - 2x^2$$

Let $h(x) = f(x) - g(x)$

$$h(x) = x^3 + 7x^2 - 5 - (5 - x - 2x^2)$$

$$h(x) = x^3 + 7x^2 - 5 - 5 + x + 2x^2$$

$$h(x) = x^3 + 9x^2 + x - 10$$

$$1) \quad h(0) = 0 + 0 + 0 - 10 = -10 < 0$$

$$2) \quad h(1) = 1 + 9 + 1 - 10 = 1 > 0$$

From ①② $\longrightarrow$ By I.V.T

Since $h(x)$ is continuous on $\mathbb{R}$ [polynomial of $d=3$]

then $h(x)$ is continuous on $[0, 1]$

then there exist at least one $c \in (0, 1)$

such that $h(c) = 0$

$$f(c) - g(c) = 0$$

$$f(c) = g(c)$$

Let $f(x) = 2^x - 1$ and $g(x) = x^3 - 1$

Use Intermediate Value Theorem to show that the graphs of f and g intersect at least once.

Past exam

Let $h(x) = f(x) - g(x)$
 $h(x) = 2^x - 1 - (x^3 - 1)$

$$h(x) = 2^x - 1 - x^3 + 1$$

$$h(x) = 2^x - x^3$$

$$h(0) = 2^0 - 0 = 1 > 0$$

$$1) \quad h(1) = 2^1 - 1 = 1 > 0$$

$$2) \quad h(2) = 2^2 - 8 = 4 - 8 = -4 < 0$$

From ① ② $\longrightarrow$ By I.V.T

Since $h(x)$ is continuous on $\mathbb{R}$ [2^x : exponential fun.
 x^3 : polynomial of deg 3]
then $h(x)$ is continuous on $[1, 2]$

then there exist at least one $c \in (1, 2)$

such that

$$h(c) = 0$$

$$f(c) - g(c) = 0$$

$$f(c) = g(c)$$

Let $f(x)$ be continuous function such that

$$f(x) = g(x) + x^4 - 3x - 2$$

Show that f and g are intersect.

Past exam

$$f(x) - g(x) = x^4 - 3x - 2$$

Let $h(x) = f(x) - g(x)$

$$h(x) = x^4 - 3x - 2$$

$$h(0) = 0 - 0 - 2 = -2 < 0$$

$$1) h(1) = 1 - 3 - 2 = -4 < 0$$

$$2) h(2) = (2)^4 - 6 - 2 = 16 - 8 = 8 > 0$$

From ① ② $\longrightarrow$ By I. V.T

Since $h(x)$ is continuous on $\mathbb{R}$ [polynomial of $d=4$]

then $h(x)$ is continuous on $[1, 2]$

then there exist at least one $c \in (1, 2)$

such that $h(c) = 0$

$$f(c) - g(c) = 0$$

$$f(c) = g(c)$$

Let $f(x) = 3x^5 + 2x^4 + x - 4$

Show that there is a real number c such that $f(c) = 10$

$$f(0) = 0 + 0 + 0 - 4 = -4 < 10$$

$$1) f(1) = 3 + 2 + 1 - 4 = -2 < 10$$

$$\begin{aligned} 2) f(2) &= 3(2)^5 + 2(2)^4 + 2 - 4 \\ &= 3(32) + 2(16) - 2 \\ &= 96 + 32 - 2 = 126 > 10 \end{aligned}$$

From ① ② $\longrightarrow$ By I.V.T

Since $f(x)$ is continuous on $\mathbb{R}$ [polynomial of degree 5]

then $f(x)$ is continuous on $[1, 2]$

then there exist at least one $c \in (1, 2)$

such that $f(c) = 10$



إذا قال أسب ان $f(c) = 10$
للزم النواتج الكبروا منه 10

Let $f(x) = g(x) + 5x$

If $g(0) = -1$, $g(1) = 0$.

f and g are continuous on $[0, 1]$.

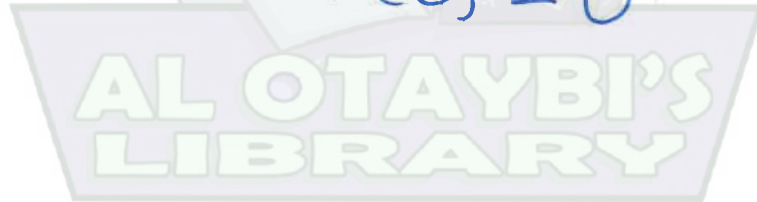
Show that f has at least one real root.

$$1) f(0) = g(0) + 0 = -1 < 0$$

$$2) f(1) = g(1) + 5 = 0 + 5 = 5 > 0$$

From ① ② $\longrightarrow$ By I.V.T

Since f is continuous on $[0, 1]$
then there exist at least one $c \in (0, 1)$
such that $f(c) = 0$



Show that the equation h has at least one real root

$$h(x) = f(x) - 5g(x) + 5$$

Where $f(0) = g(1) = 1$ and $f(1) = g(0) = 2$

, f and g are continuous on $[0, 1]$

$$h(x) = f(x) - 5g(x) + 5$$

$$\begin{aligned} 1) \quad h(0) &= f(0) - 5g(0) + 5 \\ &= 1 - 5(2) + 5 = 1 - 10 + 5 = -4 < 0 \end{aligned}$$

$$\begin{aligned} 2) \quad h(1) &= f(1) - 5g(1) + 5 \\ &= 2 - 5(1) + 5 = 2 > 0 \end{aligned}$$

From ① ② $\longrightarrow$ By I.V.T

since f and g are continuous on $[0, 1]$

then $h(x)$ is continuous on $[0, 1]$

then there exist at least one $c \in (0, 1)$

such that $h(c) = 0$

Suppose that the function f is continuous on $[0, 1]$ and satisfies $0 < f(x) < 1$ for all x in $[0, 1]$. Use the Intermediate Value Theorem to show that the equation $f(x) - x = 0$ has a solution in $(0, 1)$.

Mar 2018

Let $h(x) = f(x) - x$

$$1) h(0) = f(0) - 0 = f(0) > 0$$

$$2) h(1) = f(1) - 1 < 0 \quad \text{Since } f(1) < 1$$

From ① ② $\longrightarrow$ By I.V.T

Since $h(x)$ is continuous on $[0, 1]$

then there exist at least one $c \in (0, 1)$

such that

$$h(c) = 0$$

$$f(c) - c = 0$$

$$f(c) = c$$

Given that f and g are two continuous functions on $(-\infty, \infty)$ such that $f(0) = g(1) = 0$ and $f(1) = g(0) = 1$

Show that $\exists c \in (0, 1)$ such that $f(c) = 10g(c)$

Let $h(x) = f(x) - 10g(x)$

$$\begin{aligned} f(c) - 10g(c) &= 0 \\ f(x) - 10g(x) &= 0 \\ \Rightarrow h(x) &= f(x) - 10g(x) \end{aligned}$$



$$1) h(0) = f(0) - 10g(0) = 0 - 10(1) = -10 < 0$$

$$2) h(1) = f(1) - 10g(1) = 1 - 10(0) = 1 > 0$$

From ① ② $\longrightarrow \therefore$ By I.V.T

Since f and g are continuous on $\mathbb{R}$

then h is continuous on $\mathbb{R}$

then h is continuous on $[0, 1]$

then there exist at least one $c \in (0, 1)$

such that $h(c) = 0$

$$f(c) - 10g(c) = 0$$

$$f(c) = 10g(c)$$



Continuity on interval



الاتصال بفترة

$[a, b]$

إذا كانت مفتوحة
لا يتم تطبيق الخطوة الثانية

إذا كانت مفتوحة
لا يتم تطبيق الخطوة الثالثة

خطوات الحل

(1) إيجاد مجال الدالة , والتأكد من أن الفترة المفتوحة داخل المجال.

(2) صورة البداية = اللمت لها من اليمين.

$$f(a) = \lim_{x \rightarrow a^+} f(x)$$

(3) صورة النهاية = اللمت لها من اليسار.

$$f(b) = \lim_{x \rightarrow b^-} f(x)$$

Check for continuity of $f(x)$ on the given interval

$$f(x) = \sqrt{x-4} \quad , \quad [4, 8]$$

1) Domain of $f(x)$: $x-4 \geq 0 \rightarrow x \geq 4 \rightarrow \therefore D_f = [4, \infty)$

$$\therefore (4, 8) \subseteq [4, \infty)$$

$\therefore f$ is continuous on $(4, 8)$

Let $a \in (4, 8)$

$$f(a) = \sqrt{a-4}$$

$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} \sqrt{x-4} = \sqrt{a-4}$$

$$\therefore f(a) = \lim_{x \rightarrow a} f(x)$$

$$2) f(4) = \sqrt{4-4} = 0$$

$$\lim_{x \rightarrow 4^+} \sqrt{x-4} = 0$$

$$\Rightarrow \therefore f(4) = \lim_{x \rightarrow 4^+} f(x)$$

$$3) f(8) = \sqrt{8-4} = \sqrt{4} = 2$$

$$\lim_{x \rightarrow 8^-} \sqrt{x-4} = \sqrt{8-4} = \sqrt{4} = 2$$

$$\Rightarrow \therefore f(8) = \lim_{x \rightarrow 8^-} f(x)$$

From ① ② ③

$\therefore f(x)$ is continuous on $[4, 8]$



Text book question



Use the definition of continuity and the properties of limits to show that the function is continuous on the given interval

$$f(x) = x + \sqrt{x-4}, [4, \infty)$$

1) Domain of $f(x)$: $x-4 \geq 0 \longrightarrow x \geq 4 \longrightarrow \therefore D_f = [4, \infty)$

$$\therefore (4, \infty) \subseteq [4, \infty)$$

$\therefore f$ is continuous on $(4, \infty)$

Let $a \in (4, \infty)$

$$f(a) = a + \sqrt{a-4}$$

$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} (x + \sqrt{x-4})$$

$$= a + \sqrt{a-4}$$

$$\therefore f(a) = \lim_{x \rightarrow a} f(x)$$

2) $f(4) = 4 + \sqrt{4-4} = 4 + 0 = 4$
 $\lim_{x \rightarrow 4^+} [x + \sqrt{x-4}] = 4 + 0 = 4 \longrightarrow \therefore f(4) = \lim_{x \rightarrow 4^+} f(x)$

From ① ② $\therefore f(x)$ is continuous on $[4, \infty)$

Use the definition of continuity and the properties of limits to show that the function is continuous on the given interval

$$g(x) = \frac{x-1}{3x+6}, (-\infty, -2)$$

1) Domain of $g(x)$: $3x+6 \neq 0 \longrightarrow 3x \neq -6 \longrightarrow x \neq -2$
 $\implies \therefore D_g = \mathbb{R} \setminus \{-2\}$

$$\therefore (-\infty, -2) \subseteq \mathbb{R} \setminus \{-2\}$$

$\therefore g$ is continuous on $(-\infty, -2)$

Let $a \in (-\infty, -2)$

$$f(a) = \frac{a-1}{3a+6}$$

$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} \frac{x-1}{3x+6} = \frac{a-1}{3a+6}$$

$$\therefore f(a) = \lim_{x \rightarrow a} f(x)$$